Power & Enclosure Design Decision Report

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1. Introduction

The power and enclosure workstream is one of three project workstreams part of the research phase of the Local Hazard Network project. It is tasked to research the implementation of a system that delivers power to sensor and network devices, and a system that mounts and houses those devices.

2. Objectives

This report aims to provide a strong foundation for power and enclosure project decisions in the future, as well as recommend design solutions that should be taken forward into the next project stages. Additionally, this report is key to requirements and criteria tracking, and general project documentation.

3. Method

1. Requirements analysis
2. Discuss previous project design
3. Analyse defined hardware
4. Power system design discussion
5. Enclosure system design discussion
6. Final recommendations and considerations

4. Preliminary Relevant Requirements

Table 1. Power and Enclosure Requirements

ID	**Requirement**
R1	Each node cost shall be under AUD$1000
Power	
R2	Each node shall last 7 days without direct sunlight with a full battery
R3	Each node shall be able to reboot successfully after power loss
R4	Energy generation shall be greater than energy consumption on average
R5	All components shall be electrically compatible with each other
Enclosure	
R6	Each enclosure shall prevent particle and water ingress
R7	All components shall be accessible when the enclosure is opened
R8	Each enclosure shall fit all internal components
R9	Each node shall be able to be moved by one person
R10	Each node shall be able to be installed by one person
R11	Each node shall have its purpose clearly labelled to inform passersby of their function
R12	Each node shall be able to be deployed without significant technical expertise

5. 2025 Project Power & Enclosure

For context, the 2025 team had the following architecture for their flood monitoring sensor: 40W Solar Panel → 20A Kings MPPT Charge Controller → 12V 6.5Ah Sealed Lead-Acid (SLA) Battery → 5V USB output → MCU + Radio. The enclosure was composed of a COTS waterproof black box, internal carpet for component protection, cable glands for external wiring, L-bracket to mount on a 40x40mm SHS pole, and a solar panel on a custom bracket at a ~29-degree tilt.

5.1 Rain Guage Node (inherited from 2024 project)

- Generation: 40W panel (12Vnom, 18.2Vmpp, 2.19Ampp)
- Storage: 6.5Ah, 12V, SLA battery
- Conditioning: 12V, 5A PWM solar charge controller (5V, 2A USB output)
- Enclosure: IP65, material unknown
- Mounting: Customisable galvanised steel bracket, steek L bracket, steel screws

5.2 Creek-level Node (inherited from 2024 project)

- Generation: 40W panel (12Vnom, 18.2Vmpp, 2.19Ampp)
- Storage: 9Ah, 12V, SLA battery
- Conditioning: 12V, 5A PWM solar charge controller (5V, 2A USB output)
- Enclosure: IP65, material unknown
- Mounting: steel L bracket, steel screws, steel flat plate

5.3 Repeater Node (designed in 2025 project)

- Generation: 5W integrated panel (floodlight)
- Storage: 3.6Ah integrated battery (floodlight) chemistry unknown (or maybe 18650 Li-Ion but it's unclear)
- Conditioning: Integrated conditioning (floodlight)
- Enclosure: 3D-printed ABS, silicone sealant
- Mounting: Steel screws, adjustable 3D-printed bracket

6. Power Consumption and Hardware Constraints

6.1 Power Consumption Preliminaries

To determine what components are needed for the sensor node, the following equations and concepts must be explained first.

Power (measured in Watts (W)) is equal to voltage (measured in Volts (V)) multiplied by current (measured in Amps (A)): $P = V \times I$. While power is the rate of energy at any instant, energy is power accumulated over time: $E = P \times t$. In battery systems, energy is expressed in milliampere-hours (mAh), so a 1000mAh battery could theoretically supply 1000mA for 1 hour, or 100mA for 10 hours, etc.

Duty cycling is the concept of the node spending most it's time asleep, and waking up briefly to take readings and transmit, and going back to sleep. The average current draw is calculated as:

$$I_{avg} = \left(I_{active} \times t_{active} + I_{sleep} \times t_{sleep}\right) \div t_{total} \quad (1)$$

To determine our battery capacity, we introduce a safety factor to account for battery aging, temperature effects, and inefficiencies. A typical industry-standard safety factor for battery sizing is a margin of 25% (Test 2020). To calculate our battery capacity:

$$t_{backup} = \frac{Battery\ Capacity\ (mAh) \times\ Safety\ Factor}{I_{avg}\ (mA)} \quad (2)$$

$$Battery\ Capacity\ (mAh) = \frac{t_{backup} \times I_{avg}(mA)}{Safety\ Factor} \quad (3)$$

To determine our minimum required solar panel power (wattage), the following equation is used:

$$\frac{Battery\ Capacity(mAh) \times Nominal\ Battery\ Voltage(V)}{1000\ \times\ Peak\ Sun\ Hours\ \left(\frac{hours}{day}\right)} \quad (4)$$

6.2 Hardware Constraints

This year, our project will utilise the RAK WisBlock platform instead of the ESP32-C3 + Heltec V3 (IoT development board, WiFi LoRa, used for meshtastic) combination. RAK Wisblock is a modular open-source IoT platform designed for rapid development of low-power, wide-area network systems (RAKwireless 2026). Our project will utilise the RAK19007 as the base, and the RAK4630 as the core (Microcontroller Unit (MCU) and Radio).

6.3 Power Consumption Estimates

6.3.1 RAK19007 (Board)

Table 2. RAK19007 Board Power Consumption

Condition	Current Average
Leakage current, no modules attached	2 µA
Sleep (MCU + sensor in sleep mode)	10 µA
Active (LoRa module transmitting)	130 mA

6.3.2 RAK4630 (MCU and Radio)

Table 3. RAK4630 Power Consumption

Modes	Current Average	Voltage Required
Sleep	4.42 µA	3.3
LoRaWAN Sleep	4.23 µA	3.3
Idle	3.35 mA	3.3
RX Window	2.22 mA	3.6
TX (15 Bytes)	67.8 mA	3.6

6.3.3 Modes

Sleep mode is when the onboard MCU has shut down most of its functions and has entered a deep sleep state. This chip has gone to sleep and is awoken on an interrupt or timer. **LoRaWAN sleep** is like sleep but measured within the context of running LoRaWAN firmware. **Idle mode** is the state when the MCU is fully powered and running but not doing any active work. **RX window** is when the LoRa radio is actively listening for incoming messages. **TX sending** is when the LoRa radio is transmitting a packet (15 bytes). The RX window and TX sending values are based on the RAK500-O which is like the RAK19007.

6.4 Sampling Period Recommendation

Based on research on similar sensors systems such as New York's FloodNet (Mydlarz et al., 2024) and a paper on using low-power sensor nodes for environmental monitoring (Reis, 2026), we can expect transmitting time to be around 3 seconds.

Wade, one of our external experts, mentions that measurement frequency will depend on the hazard being monitored, and during high-risk periods. For example, flood sensors may transmit data every 12 hours as the trend if water is rising or falling matters more than real-time data. During flooding events, frequency may increase up to 30 minutes during the day and less frequently during extended cloud cover to conserve power.

For a conservative estimate, our sensors will transmit every 30 minutes, increasing to up to every minute depending on the hazard and risk period.

7. Energy Generation

Our sensor systems will be placed in remote, and hard to access areas, on-site energy generation is essential. All previous iterations of this project used a solar panel for this purpose, and it is common practice for outdoors, standalone devices.

7.1 Photovoltaics Preliminaries

The most common solar panels (monocrystalline silicon) use a p-n junction semiconductor material to separate electron-"hole" pairs created by photons exciting valence electrons, producing a current in the connected circuit. Solar panels are made up of solar modules, which are made up of solar cells. The combination of cells and modules in series and parallel control the magnitude of voltage and amperage values. Other semiconductors and constructions use the same photovoltaic effect, with some differences in absorption, efficiency, degradation, voltage and current, and cost but are still beaten out by Silicon panels for most consumer-level applications (Mdallal et al., 2025).

7.2 Solar Resource and Site Analysis

Solocast (Solocast, 2025) provides data on Global Horizontal Irradiance (GHI, total amount of solar radiation received by a horizonal surface, W/m^2) radiance over a year. This value was converted to Peak Sun Hours (PSH, the equivalent number of hours per day when solar irradiance averages 1000 watts per square meter). For our four locations, data was taken over the year for 2025. Figure 1 shows that Terania, Uki and Main Arm have around the same number of PSH per day over the course of the year, reaching the lowest from April to July. Mandagery generally has more extremes having more PSH in the summer, and lower PSH in the winter than the other location.

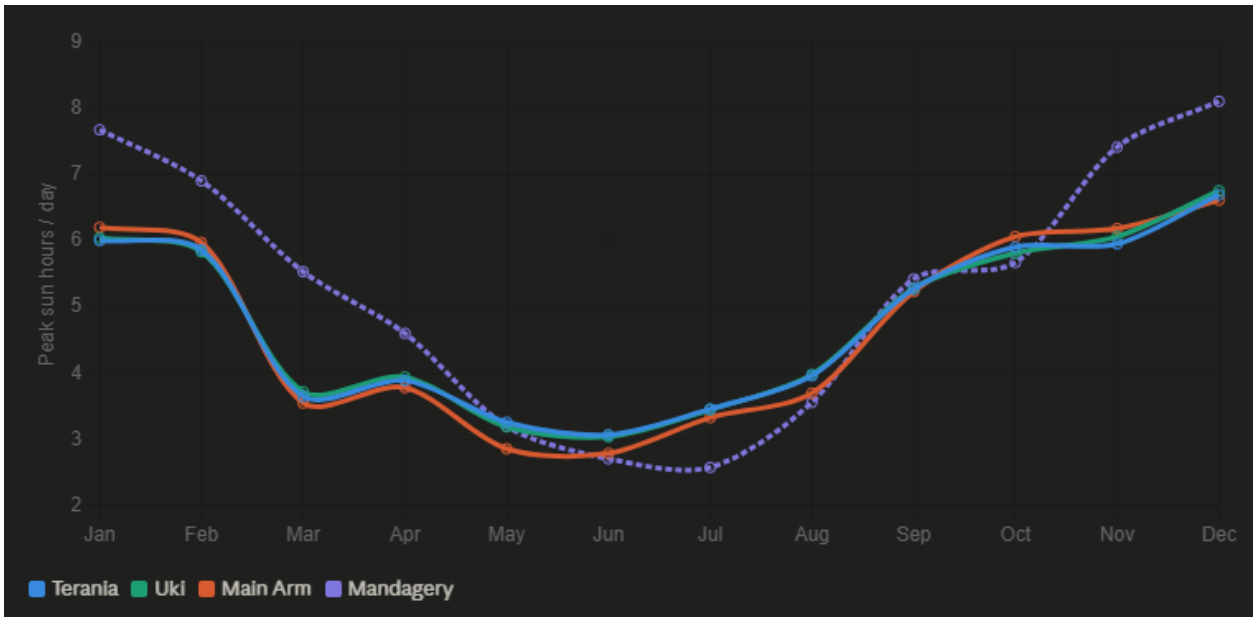


Figure 1. Peak Sun Hours per Day across focus locations for 2025

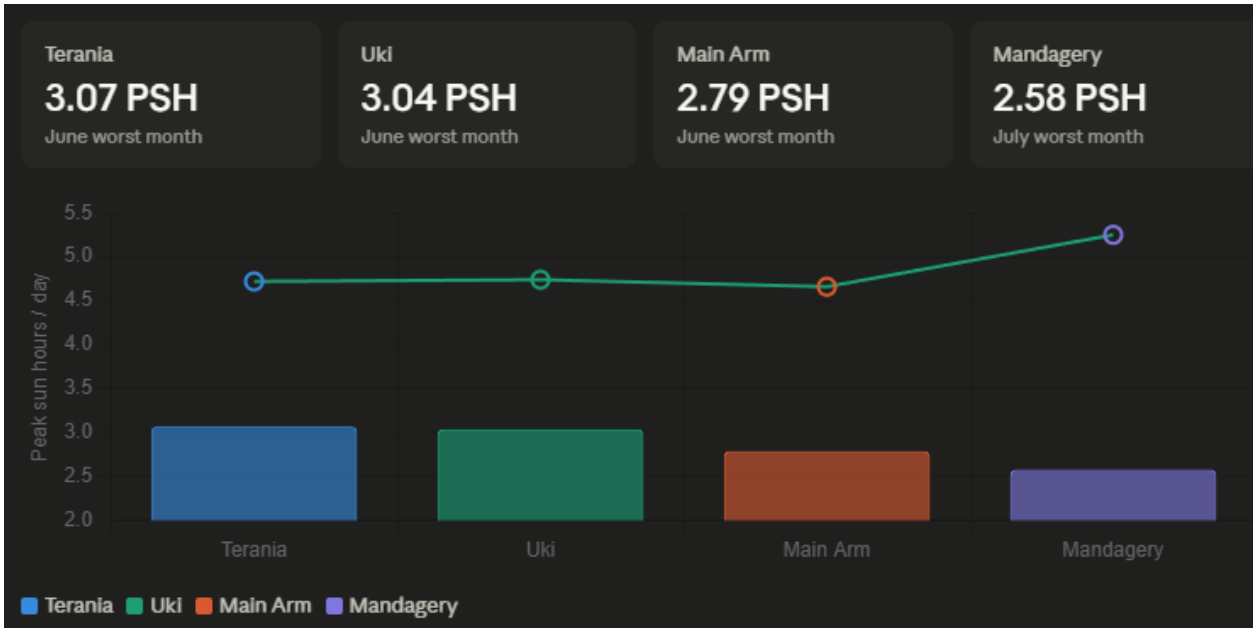


Figure 2. Dot - Average Annual PSH, Bar - Worst Month PSH

Data sourced from Solocast. Data collation and visualisations by Claude. Interactive visualisation available at: https://claude.ai/public/artifacts/792cc7e1-7454-4435-b3ec-e69a7cef6f92. Raw data available in resources folder in repository.

Figure 2 shows that worst case over the year is Mandagery at 2.58 PSH in July. We should aim to ensure our sensors and repeaters are able to stay online during periods of very little sunlight.

7.3 Solar Panel

Monocrystalline silicon solar panels are the most common consumer construction due to their high relative conversion efficiency (highest recorded was 26.1%), robustness, and low cost (0.11 $/kWh) (Mdallal et al., 2025). Specific output voltage and current vary by panel and size, but the range available allows us to choose one closest to what we need.

Several environmental factors significantly impact power output and degradation, which is especially relevant for our product that should remain operational during periods without sun, and with increased temperature.

Cell temperature increases during operation, stabilising at a point that depends on the ambient temperature, where increased temperatures reduce panel power output (Hoymiles, 2022).

The second important factor is shading of the panel, which decreases capturable power, but also can increase cell temperature if the panel is partially shaded (Hoymiles, 2022).

Thirdly, panel tilt and orientation directly affect the incident sunlight across the day. Panel orientation simply needs to be facing North (Southern hemisphere) or South (Northern hemisphere. Optimal panel tilt considers the path of the sun over the year and how it changes over a year, based on the location's latitude. The tilt angle can be adjusted up from the optimal to bias for winter and adjusted down to bias for summer (Hoymiles, 2022).

Lastly, other factors that affect power output include local terrain, its albedo, the ratio of direct and diffuse sunlight, wind speed, and photon energy. These factors can be important in extreme and heavily optimised applications (Hoymiles, 2022).

Solar panel systems are usually designed with and generation factor of safety (FoS), which is the ratio of installed rated power to maximum power consumption. The recommended rated power is determined by using the worst average month of generation and consumption (worst-case method). Typical generation FoS is around 1.2 but can range from 1.1-1.4 depending on consumption behaviour (Stapleton, 2019).

7.3.1 Recommended Components

A 5W solar panel can be connected directly to the RAK19007 board through a P1 connector (designed for JST ZHR-2 1.5mm pitch female connectors). The panel output voltage must not exceed 5.5V. Exceeding this could permanently damage the board.

For a 5V solar panel with a minimum 1.2W:

Table 4. Solar Panel Recommended Components

Required Current Output	**Product**	**Link**	**Cost (AUD)**	**Notes**
90-250mA	Solar Panel	https://store.rakwireless.com/products/solar-panel	$3.50	• Size 80x45x2mm • Current at nominal power 90mA (±5%) • Cell efficiency 18% • UV resistant • Lifetime 5 years
250-380mA	RAKwireless Solar Panel with JST 1.5 connector (Large 5.5 x 3.5 inch)	https://store.rokland.com/products/rakwireless-solar-panel-with-jst-1-5-connector-large-5-5-x-3-5-inch-sku-920433	$19.00	• Size 140x90x2mm (5.5 x 3.5 inch) • Current at nominal power 380mA (±5%) • Cell efficiency 18% • UV resistant • Lifetime 5 years

8. Balance of Systems

8.1 Energy Storage (Batteries)

Reusable electrical energy storage in the form of rechargeable (or secondary) batteries stores electrical energy as a potential (voltage) between the electrodes. The anode (positive) and cathode (negative) are separated by electrolyte medium where dissociated chemicals, as ions, deliver electrons between the contacts. To be reuseable, reversing the direction of current with a voltage greater than the maximum potential in the battery, returns the ions to their original electrode, returning the stored chemical energy (Schumm, 2026b).

Batteries' chemical composition determines characteristics and effects such as:

- Nominal voltage
- Final discharge voltage
- Minimum/maximum current discharge
- Energy density (gravimetric/volumetric)
- Charge cycles
- Temperature effects
- Memory effect
- Cost per unit energy capacity (Schumm, 2026a)

The most common secondary battery chemical compositions are sealed lead-acid (SLA), lithium-ion (Li-Ion), lithium-polymer (Li-Po), nickel-metal hydride (NiMH), and nickel-cadmium (NiCd) (Schumm, 2026a).

8.1.1 Project Application

The RAK19007 base board is only compatible with a rechargeable Lithium-Ion (Li-Ion) battery. The battery must match the following values:

Table 5. Battery requirements

Parameter	**Value**
Standard voltage	3.7V
Charging voltage	2.4V
Discharge current	>500mA

Li-Ion batteries have increased energy density, charge cycles, charge rate, decreased self-discharge, and no memory effect. Their main drawbacks are their up-front cost and higher chance of thermal runaway, but they are mitigated by lifetime value and smart thermal management respectively (Bioenno Tech LLC, 2023).

Due to being constrained to use an 18650 (or similar Li-Ion) battery, the system should accommodate for where is falls short regarding requirements. It meets the

requirements of portability and low maintenance; however, the risk of thermal runaway could be an issue if there are steep instant discharges.

8.1.2 Charging Connections

The battery can be connected to the RAK19007 board via the P2 connector or USB-C. P2 is preferred as the USB-C port is mainly used for uploading firmware to the board. The following two tables outline the P2 connector and USB-C requirements.

Table 6. P2 Connector Requirements

Minimum	Typical	Maximum	Unit
3.3	3.7	4.3	V

Table 7. USB-C Requirements

Parameter	Value
Input charging voltage	4.5-5.5V
Charging current	350mA

8.1.3 Recommended Components

For batteries with a voltage of 3.7V:

Table 8. Recommended Batteries

Required Battery Capacity	Product	Link	Cost (AUD)	Notes
<400mAh	Polymer Lithium Ion Battery (LiPo) 3.7V 400mAh	https://core-electronics.com.au/polymer-lithium-ion-battery-400mah-38456.html	$5.55	• Standard 2-pin JST-PH connector • 38x25x6mm • 11g • 2C continuous discharge • Great energy retention with low self-discharge rates (<8% per month) • Reliable under extreme conditions (-10 to 50C)
400mAh – 2000mAh	Polymer Lithium Ion Battery (LiPo) 3.7V 1100mAh	https://core-electronics.com.au/polymer-lithium-ion-battery-1000mah-38458.html	$8.65	• Standard 2-pin JST-PH connector • 51x34x6mm • 21g • Rated for 2C continuous discharge • Great energy retention with low self-discharge rates (<8% per month)

				• Reliable under extreme conditions (-10 to 50C)
1100mAh-2000mAh	Polymer Lithium Ion Battery (LiPo) 3.7V 2000mAh	https://core-electronics.com.au/polymer-lithium-ion-battery-2000mah-38459.html	$14.45	• 56x56x5.8mm • 40g • 2C continuous discharge • Great energy retention with low self-discharge rates (<8% per month) • Reliable under extreme conditions (-10 to 50C)

8.2 Power Conditioning

The previous project used a Kings 20A MPPT as they used a 40W solar panel which would have outputted a voltage too high for their microcontroller and battery. As we will be using a 5V solar panel with the RAK19007, the RAK board has a charger chip that will be able to accept the solar input directly at 5V and manage the battery charging automatically (as indicated by on-board LEDs). Therefore, there will be no need for any extra components for power conditioning.

8.2.1 Additional Voltage Step-Down/Step-Up

The RAK board will not need a step-down converter as the RAK19007 board accepts up to 5.5V directly from the solar panel and we will most likely be using a 5V panel (see Solar Panel). The board only accepts up to a 4.3V battery and will not require a step-down converter.

It is likely that a step-up converter will not be needed for the sensors, The RAK19007 board has a 3.3V rail which will support sensors that operate at this voltage. If a sensor requires more than 3.3V (potentially ultrasonic distance sensor) would require a boost converter from 3.3V to 5V.

8.3 Sensor Recovery

An important consideration is that if a node goes off during bad weather, it must be able to self-heal as per Requirement R3.

The RAK19007 has a reset pin (pin 13) which can be utilised to reset the attached microcontroller. For this, an external watchdog should be used such as the TPL5010 which can connect with the RAK19007 and pulling the reset pin low to restart the microcontroller.

Testing will have to be done to see how specifically the RAK19007 deals with coming back online after a drained battery. This forum discovered that while testing with a RAK19007, there was a hardware bug which caused sensors to draw current during low-battery boot phase preventing startup, and when the battery is too low, the nRF52840

chip wakes up and tries to transmit via LoRa and shuts down indefinitely (RAKwireless Technology Limited, 2024). One of the forum contributors mention a firmware solution to potentially solve this issue.

However, an external watchdog controller may still be needed if the MCU itself fails to turn on or a full power cycle is needed. Again, testing needs to be undertaken to examine the behaviour of the sensor node after a period of no battery and solar power.

Another note is the nRF52840 chip has a built-in watchdog which can handle hardware hangs.

8.3.1 Recommended Components

If an external watchdog IC is needed:

Table 9. Recommended External Watchdogs

Product	Link	Cost (AUD)	Notes
595-TPL5010DDCT	https://au.mouser.com/ProductDetail/Texas-Instruments/TPL5010DDCT?qs=beZBGHSk9c06Mf4cbaYiUA%3D%3D	$1.57	• Supply voltage min: 1.8V • Supply voltage max: 5.5V • Operating Temp: -40C-105C • 35nA
TPS3435	https://au.mouser.com/ProductDetail/Texas-Instruments/TPS3435BAACADDFRQ1?qs=sGAEpiMZZMutXGli8Ay4kKcGz1bJHtT1dLXHxgNnVa8%3D	$5.56	• Supply voltage min: 1.04V • Supply voltage max: 6V • Operating Temp: -45C-125C • 250nA

8.4 Wiring & Internal Layout

Circuit wiring should use insulated tinned copper according to maximum current through the branch. Solar panels often come with their own lead or connectors but may need to be swapped dependent on what they are. Connectors should be insulated and wrapped with heat shrink if any gaps are left. Fuses should be implemented on between components if the peak current draw exceeds the safe limit for the source, or to isolate components in the case of a short-circuit.

To organise wiring, a bread board and prototyping board should be used in the testing and final product stages respectively. Components should be spaced from each other,

with an insulator like foam. The exact spacing depends on the available space inside the enclosure. Ideally the insulator also thermally separates components. If during testing it is found that a component's temperature is above a safe level for itself or nearby components, then a heatsink should be used to disperse heat away from those itself and those components. This could potentially inform specific layout decision as well.

8.5 Enclosure

The enclosure functions to separate the device from all external factors, except for the sensor. It should prevent dust, water, and heat from damaging everything inside. An enclosure's Ingress Protection (IP) rating describes what level of dust (first digit of rating) and water (second digit of rating) ingress it prevents. Standard ratings for outdoor enclosures for electronics are IP65, IP66, IP67 and IP68. A 6 dust-rating means its dust-tight, and the water-ratings increase from protection against water jets, to protection against continuous submersion (in the case of a flood sensor). A silicone sealant can provide additional dust and water protection. Protection against heat typically comes in the form of an insulator inside the enclosure, such as foam.

The enclosure should be made of a material that is resistant to degradation caused by UV light, meaning that many plastics and polymers can't be used unless they've undergone a specific treatment to prevent breakdown from UV. Metal enclosures are typically aluminium, which hold very little heat and don't easily corrode, and stainless steel, which are the most durable, although both are more durable than plastics. There is the additional weight trade off depending on which enclosure material is chosen. The final structural enclosure consideration is the mounting and carry options that are built in, which would remove the need for any modifications that could harm protection integrity.

8.5.1 Recommended Components

The below options are subject to the sizing and positioning requirements of components that are not yet decided.

Table 10. Recommended Enclosures

Product	Link	Cost (AUD)	Notes
Black Waterproof ABS Plastic Case	https://www.jaycar.com.au/black-waterproof-abs-plastic-case-182-x-120-x-75mm/p/HB6423	$29.95 (minus staff discount ~50%)	• 182x120x75mm • IPX8 (not tested against dust but highly rated against water) • ABS • Foam insulation but not cut-insert

			• Handles but no screw holes
ABS Instrument Case with Purge Valve	https://www.jaycar.com.au/abs-instrument-case-with-purge-valve-mpv1/p/HB6388	$59.95 (minus staff discount ~50%)	• 186x75x123mm • No IP rating • ABS • Foam cut-insert • Handle and 2 holes for fastening shut or mounting
Tactix Tough Case	https://www.bunnings.com.au/tactix-tough-case-in-black-large_p0469604	$49	• 383x269x159mm (quite large) • IP65 • ABS (most likely) • Also available in small (medium out-of-stock) • Foam cut-insert insulated • Previous project used a similar product
Sealed Diecast Aluminium Box	https://www.jaycar.com.au/ip65-sealed-diecast-aluminium-boxes-flanged-115-w-x90-d-x55-h-mm/p/HB5044	$27.95 (minus staff discount ~50%)	• 115x90x55mm • IP65 • Aluminum • Some different sizes • No insulation (should be bought separately) • Screws holes for closing and flanges with screw holes for mounting

8.6 Mounting & Installation

The method of mounting for the enclosure and external items, such as the solar panel and sensors, should be stable under potentially harsh weather conditions, with multiple points of contact for stability. It should also be robust enough to not come loose, degrade, or break. Besides the mounting method, the stability relies on the

structure it's mounted, so the structure should provide a stable base for whatever mounting method is chosen.

The requirements of the sensors, and power system restrain the potential mounting locations, so the mounting method should be adjustable enough to be installed in a variety of ways, to fit what is available at the location. Potential locations include posts, fences, trees, buildings, and bridges.

For both the prototype and final product, installation should only require one person to set up, including adjusting the mounting to fit the structure.

8.6.1 Recommended Methods & Components

Table 11. Recommended Prototyping Components

Method/Product	Description/Link	Cost (AUD)	Notes
General	To connect enclosure and external items to a secure base		
Steel flat-plate make-a-bracket	https://www.bunnings.com.au/grunt-25mm-x-2m-pull-tie-down-quick-release-light-duty_p4310165	$3.30	• 40x200x2mm • Galvanised steel • Adjustable • Tool required
Square aluminium tube (support post)	https://www.bunnings.com.au/metal-mate-25-4-x-25-4-x-1mm-x-1m-black-powder-coated-aluminium-square-tube_p0920872	$17.36	• 25.4x25.4x1000mm • 1mm thick
Strap	Use for uneven/large radius mounting structure		
Pull tie-down strap	https://www.bunnings.com.au/grunt-25mm-x-2m-pull-tie-down-quick-release-light-duty_p4310165	$5.90	• 25x2000mm • Rated for 100kg • UV-protected polyester • Weather resistant
Clamp (no drill holes)	Use for even/small radius mounting structure		
Steel flat-plate make-a-bracket	https://www.bunnings.com.au/carinya-40-x-200-x-2mm-galvanised-flat-plate-make-a-bracket_p3960602	$3.30	• 40x200x2mm • Galvanised steel • Adjustable

			• Tool required
Flange-screwed (drilling into structure)	Use for flat surface		
Steel flat-plate make-a-bracket	https://www.bunnings.com.au/carinya-40-x-200-x-2mm-galvanised-flat-plate-make-a-bracket_p3960602	$3.30	• 40x200x2mm • Galvanised steel • Adjustable • Tool required

The solar panel and external sensors should be mounted to a support post or plate, and the solar panel should be mounted on a roughly 30 degree angle, facing North.

9. Repeater Node

A repeater listens for messages from devices and other repeaters and rebroadcasts those messages to extend coverage of the data (NoDak Mesh, 2026).

9.1 Similarities to Sensor Nodes

The RAK19007 and RAK4630 WisBlock combination will be used again for the repeaters which means the 5V panel and 3.7V Li-Ion battery constraints still applies. Requirement 3 self-recovery is also still important.

9.2 Differences to Sensor Nodes

The main difference will the power consumption calculation which will affect the required values for the solar panel and battery. As the repeaters should be always ready to receive a signal, they will draw more current than the sensors nodes.

10. Component Recommendations

Component recommendations from the above sections are restated here for reference.

10.1.1 Solar Panel

For a 5V solar panel with a minimum 1.2W:

Table 12. Recommended Solar Panels

Required Current Output	Product	Link	Cost (AUD)	Notes
90-250mA	Solar Panel	https://store.rakwireless.com/products/solar-panel	$3.50	• Size 80x45x2mm • Current at nominal power 90mA (±5%) • Cell efficiency 18% • UV resistant • Lifetime 5 years
250-380mA	RAKwireless Solar Panel with JST 1.5 connector (Large 5.5 x 3.5 inch)	https://store.rokland.com/products/rakwireless-solar-panel-with-jst-1-5-connector-large-5-5-x-3-5-inch-sku-920433	$19.00	• Size 140x90x2mm (5.5 x 3.5 inch) • Current at nominal power 380mA (±5%) • Cell efficiency 18% • UV resistant • Lifetime 5 years

10.1.2 Battery

For batteries with a voltage of 3.7V:

Table 13. Recommended Batteries

Required Battery Capacity	Product	Link	Cost (AUD)	Notes
<400mAh	Polymer Lithium Ion Battery (LiPo) 3.7V 400mAh	https://core-electronics.com.au/polymer-lithium-ion-battery-400mah-38456.html	$5.55	• Standard 2-pin JST-PH connector • 38x25x6mm • 11g • 2C continuous discharge • Great energy retention with low self-discharge rates (<8% per month) • Reliable under extreme conditions (-10 to 50C)
400mAh – 2000mAh	Polymer Lithium Ion Battery (LiPo) 3.7V 1100mAh	https://core-electronics.com.au/polymer-lithium-ion-battery-1000mah-38458.html	$8.65	• Standard 2-pin JST-PH connector • 51x34x6mm • 21g • Rated for 2C continuous discharge • Great energy retention with low self-discharge rates (<8% per month) • Reliable under extreme conditions (-10 to 50C)
1100mAh - 2000mAh	Polymer Lithium Ion Battery (LiPo) 3.7V 2000mAh	https://core-electronics.com.au/polymer-lithium-ion-battery-2000mah-38459.html	$14.45	• 56x56x5.8mm • 40g • 2C continuous discharge • Great energy retention with low self-discharge rates (<8% per month) • Reliable under extreme conditions (-10 to 50C)

10.1.3 External Watchdog IC

If an external watchdog IC is needed:

Table 14. Recommended External Watchdogs

Product	Link	Cost (AUD)	Notes
595-TPL5010DDCT	https://au.mouser.com/ProductDetail/Texas-Instruments/TPL5010DDCT?qs=beZBGHSk9c06Mf4cbaYiUA%3D%3D	$1.57	• Supply voltage min: 1.8V • Supply voltage max: 5.5V • Operating Temp: -40C-105C • 35nA
TPS3435	https://au.mouser.com/ProductDetail/Texas-Instruments/TPS3435BAACADDFRQ1?qs=sGAEpiMZZMutXGli8Ay4kKcGz1bJHtT1dLXHxgNnVa8%3D	$5.56	• Supply voltage min: 1.04V • Supply voltage max: 6V • Operating Temp: -45C-125C • 250nA

10.1.4 Enclosure

Table 15. Recommended Enclosures

Product	Link	Cost (AUD)	Notes
Black Waterproof ABS Plastic Case	https://www.jaycar.com.au/black-waterproof-abs-plastic-case-182-x-120-x-75mm/p/HB6423	$29.95 (minus staff discount ~50%)	• 182x120x75mm • IPX8 (not tested against dust but highly rated against water) • ABS • Foam insulation but not cut-insert • Handles but no screw holes
ABS Instrument Case with Purge Valve	https://www.jaycar.com.au/abs-instrument-case-with-purge-valve-mpv1/p/HB6388	$59.95 (minus staff discount ~50%)	• 186x75x123mm • No IP rating • ABS • Foam cut-insert • Handle and 2 holes for

			fastening shut or mounting
Tactix Tough Case	https://www.bunnings.com.au/tactix-tough-case-in-black-large_p0469604	$49	• 383x269x159mm (quite large) • IP65 • ABS (most likely) • Also available in small (medium out-of-stock) • Foam cut-insert insulated • Previous project used a similar product
Sealed Diecast Aluminium Box	https://www.jaycar.com.au/ip65-sealed-diecast-aluminium-boxes-flanged-115-w-x90-d-x55-h-mm/p/HB5044	$27.95 (minus staff discount ~50%)	• 115x90x55mm • IP65 • Aluminum • Some different sizes • No insulation (should be bought separately) • Screws holes for closing and flanges with screw holes for mounting

10.1.5 Mounting & Installation

Table 16. Recommended Prototyping Components

Method/Product	Description/Link	Cost (AUD)	Notes
General	To connect enclosure and external items to a secure base		
Steel flat-plate make-a-bracket	https://www.bunnings.com.au/grunt-25mm-x-2m-pull-tie-down-quick-release-light-duty_p4310165	$3.30 x2	• 40x200x2mm • Galvanised steel • Adjustable • Tool required
Square aluminium tube (support post)	https://www.bunnings.com.au/metal-mate-25-4-x-25-4-x-1mm-x-1m-black-powder-coated-aluminium-square-tube_p0920872	$17.36	• 25.4x25.4x1000mm • 1mm thick

Strap	Use for uneven/large radius mounting structure		
Pull tie-down strap	https://www.bunnings.com.au/grunt-25mm-x-2m-pull-tie-down-quick-release-light-duty_p4310165	$5.90	• 25x2000mm • Rated for 100kg • UV-protected polyester • Weather resistant
Clamp (no drill holes)	Use for even/small radius mounting structure		
Steel flat-plate make-a-bracket	https://www.bunnings.com.au/carinya-40-x-200-x-2mm-galvanised-flat-plate-make-a-bracket_p3960602	$3.30	• 40x200x2mm • Galvanised steel • Adjustable • Tool required
Flange-screwed (drilling into structure)	Use for flat surface		
Steel flat-plate make-a-bracket	https://www.bunnings.com.au/carinya-40-x-200-x-2mm-galvanised-flat-plate-make-a-bracket_p3960602	$3.30	• 40x200x2mm • Galvanised steel • Adjustable • Tool required

10.2 Procurement and Cost

Based on the recommended components, the expected cost of the power and enclosure portions of the sensor and repeater nodes is between AUD$65.83 to AUD$128.82. All components can be purchased online and shipped within Australia.

10.3 Functional Block Diagram

https://excalidraw.com/#json=X99VfQAbW41DL2uf_oeSr,hRNZMIvSmpM8u7DNNXGw2Q

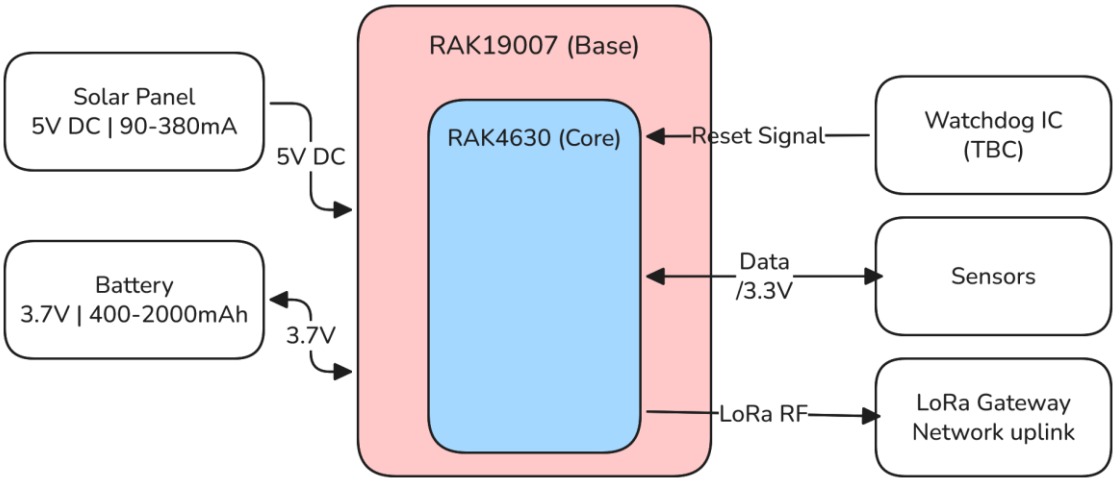


Figure 3. Preliminary Functional Block Diagram for the Sensor System

11. Future Work

11.1 Workstream Integration

The following points will require discussions with the other workstream reports which may impact some of the requirements and processes for the sensor and the repeater:

11.1.1 Network Workstream

- Re: Wade - node health metadata (battery percentage, online/offline status) is important for network management
- How long exactly to transmit and receive data
- Antennae mounting and connection to board
- How many sensors will connect to the repeaters?

11.1.2 Sensors Workstream

- How often to transmit data depending on hazard
- How sensors will be mounted

12. Conclusion

Based on the calculations and requirements discussed in this report, the following spreadsheet calculator was created to determine battery capacity, solar panel power, and solar panel current required for the sensor and repeater: Specifications, Requirements & Criteria.xlsx.

The cells highlighted in blue are calculation cells. The cells highlighted in yellow are the factors used to decide the appropriate solar panel and battery.

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